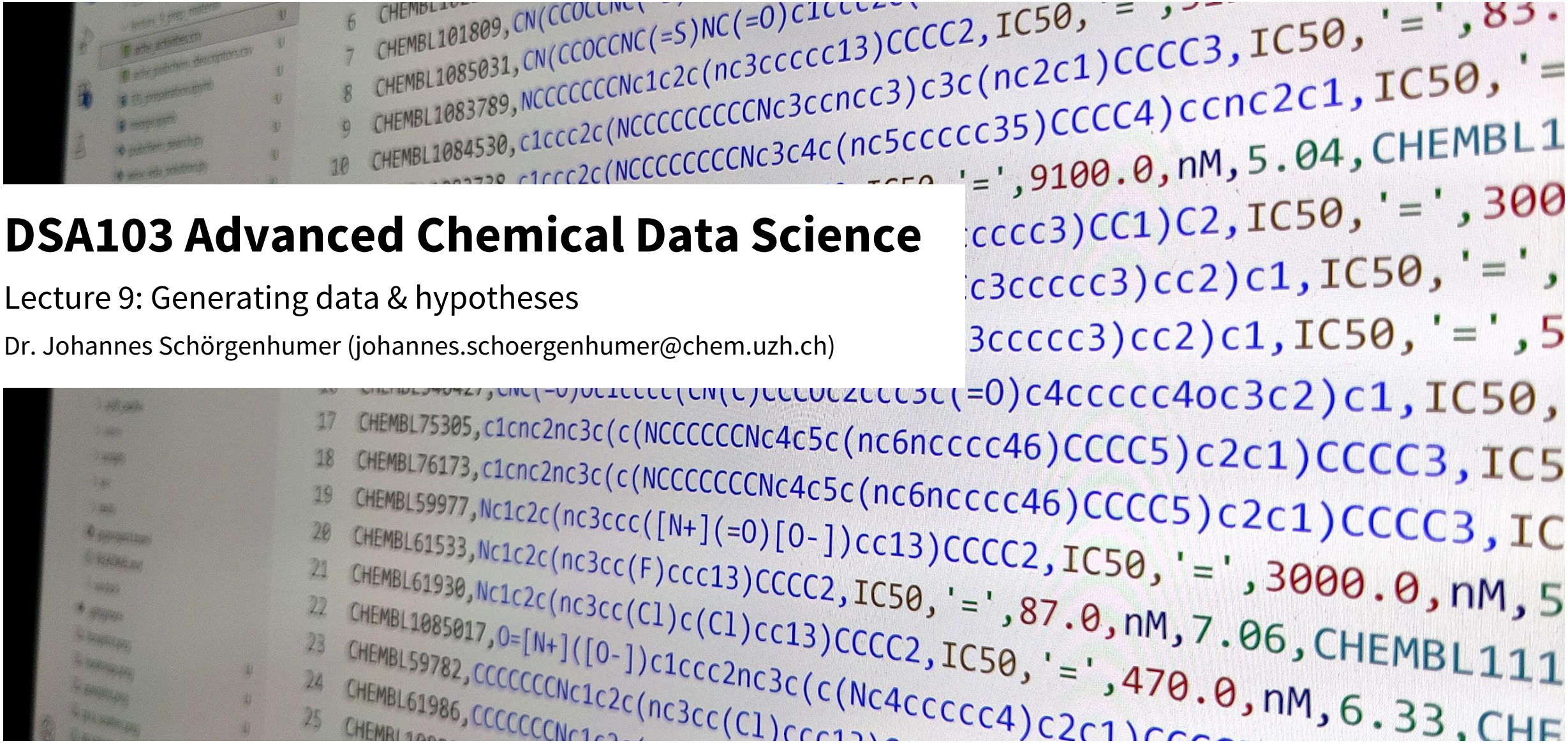
Universität Zürich[UZH]

Department of Chemistry



DSA103 Advanced Chemical Data Science

Lecture 9: Generating data & hypotheses

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Content overview

Recap: What is data good for?

Creating a dataset

How to deal with bad data

Hypothesis generation

Hypothesis testing



GUINNESS

Deciding what the data is good for

Points to consider:

- What does the data actually present?
- What is the scope and scale of the data?
- How was the data produced?
- What metadata is available?
- What kind of quesitons can the data answer?
- What about limitations and biases?
- Is the data trustworthy?
- Can this data be combined with other data sources?
- What about ethical or legal issues?
- What story could the data tell?

Acitivity: Assessing data quality and what we can do with the data

Let‘s look again at an entry in PubChem:

- What does the data show?
- Where does the data come from?

Now imagine a collection of data:

- What can (and can’t) we do with it?

Creating a data set – general process

Creating a dataset for scientific purposes is not only about collecting data, but also a **systematic**, **reliable** and **ethical** process to ensure that the data set is of high quality, relevant and useful.



Some essential steps:

1) Define **objective**: What question are we trying to answer? Which models are going to be used?
2) Identify **data requirements**: Which features? Which types of data, which formats? How much?
3) Choice of **data source**
4) Data **collection**: Digitize manual entries, assemble data collection using e.g. pandas
5) Data **cleaning and preprocessing**: handling “bad” entries, outliers, convert to uniform data format
6) Data **validation** and quality assessment: Check distributions, assess biases, visualisation!

On data sources

Data can be obtained from many different sources, depending on the scale, accessibility, purpose, …, e.g.:

- Manual data collection
- Synthetic data generation
- Simulation data
- Web scraping
- Open Datasets

In the life sciences, most of the time we are confronted with:

- (Our own) Experimental data
- Computational results
- Literature and patent data
- Open research databases

Which challenges could be associated to these?
What solutions are there?

How to deal with bad data points – duplicates

Not every duplicate is an error – but every duplicate needs to be assessed

- Data entry errors (e.g. repeated records by error) **remove**
- Systematic duplication (e.g. the same event logged twice due to server restart) **verify, remove**
- Legitimate repetition (e.g. repeated observation, same measurements) **keep**
- Data integration artifacts (e.g. merging two datasets with overlapping keys) **remove**

How to assess:

1) Check for duplicates and quantify: **df.duplicated().sum()**
2) Inspect full-row duplicates: **df[df.duplicated(keep=False)]**
3) Partial duplicates (same ID but different attributes, e.g. after an update): inspect by grouping, e.g. by “name”
4) Decide: Keep, flag, update or drop?

Good practice for dropping duplicates:

- Document why duplicates were dropped, e.g. as Markdown or comment
- Keep original dataframe: **df_cleaned = df.drop_duplicates()**

How to deal with bad data points – outliers

Generally, outliers are not bad data.

- Error outlier(e.g. physically impossible values): bad data **remove**
- Contextual outlier (e.g. valid value but out of the range in context) **Investigate and decide**
- Informative outlier (e.g. meaningful, rare event) **Really important, keep!**

Suggested cleaning workflow for outliers:

1) **Detect** (statistical or visual methods)
2) **Inspect** (is it plausible? check raw data, metadata)
3) **Decide** (keep, correct, flag, or remove)
4) **Document** (comment, markdown)

How to deal with bad data points – missing values

Pandas: **NaN** (“Not a number”, floating point value) and **None** (singleton object), handled basically interchangeably and interconverted where needed.

Detection: **isna()** (alias: **isnull()**), **notna()** (alias: **notnull()**)

How to deal with missing values:

1) **dropna()**
   - By default deletes any row that contains at least one missing value, deleting columns can be specified (**dropna(axis=1)**)
   - To introduce a threshold for how many NaN are still acceptable: **thresh=x**
   - To specify to only drop rows if all entries are missing: **how=“all”**
   - A subset of columns can be specified to only drop rows if the missing value is in one of them: **dropna(subset=[“columns”])**
2) **fillna()**
   - Fill missing value with zero (or any other number): **df.fillna(0)**
   - Missing data can also be filled with previous (**df.fillna(method='ffill’)**) or next value in the column fillna(**df.method=‘bfill'**)

Good practice: document and keep copy of original data!

https://jakevdp.github.io/PythonDataScienceHandbook/03.04-missing-values.html

Acitivity: Generating a data set

Curate dataset for a purpose **Acetylcholinesterase (AChE inhibitors – Structure-Activity-Relation (Binding assay))**

1) Define objective: What question are we trying to answer?

Acitivity: Generating a data set

Curate dataset for a purpose **Acetylcholinesterase (AChE inhibitors – Structure-Activity-Relation (Binding assay))**

1) Define objective: What question are we trying to answer? **Can we correlate molecular properties, e.g. lipophilicity with the activity in AChE inhibition assays?**
2) Identify data requirements: Which features? Which types of data, which formats? How much? **pchembl_value (activity), xlogP (lipophilicity)**
3) Choice of data source **Acitvity data (ChEMBL), molecular properties (PubChem)**
4) Data collection: Use Pandas to combine (merge using common key)
5) Data cleaning and preprocessing: handle duplicates and missing values
6) Data validation and quality assessment: Check distributions, assess biases, visualization!

Hypothesis generation

Key elements in hypothesis generation:

- **Observations & Patterns**: Starting from initial insights or after exploratory data analysis
- **Domain Knowledge**: Leveraging expertise or existing research in the field
- **Clarity**: The hypothesis should be clear and testable
- **Direction**: It provides a direction for analysis or model-building

Parts of the hypothesis:

- **Null-hypothesis (H_0):** Assumes that there are no effects or differences between the variables of two or more samples. Always tested.
- **Alternate hypothesis (H_1):** Assumes there is an effect or difference.

Level of significance: defines the distance from H_0 at which a sample parameter is statistically significant. Usually set to 0.05

P-value: probability of observed result from testing. P-value < 0.05 means H_1 is true, H_0 is rejected

Hypothesis testing

Why?

- **Validates the hypothesis**: Are observed patterns from data exploration significant or just coincidental
- **Guides decision-making**
- **Avoids biases**: Proper testing helps avoid confirmation bias: conclusions should be backed by data and not just assumptions or guesses.
- Testing should lead to accept, refine or reject the hypothesis

Tools:

- **t-test**
- A/B tests
- Chi-square test
- ANOVA
- …

https://towardsdatascience.com/hypothesis-testing-with-python-step-by-step-hands-on-tutorial-with-practical-examples-e805975ea96e/

Hypothesis testing – t-test (using scipy)



Also called *Student's t-test*: one of the most common tests in hypothesis testing.

Central question: **Is the difference between the response of two groups statistically significant**?

Originally derived by William Sealy Gosset in 1908 for assessing the quality of stout.

Based on different assumptions (e.g. random samples, data ap. normally distributed): **different tests, e.g.:**

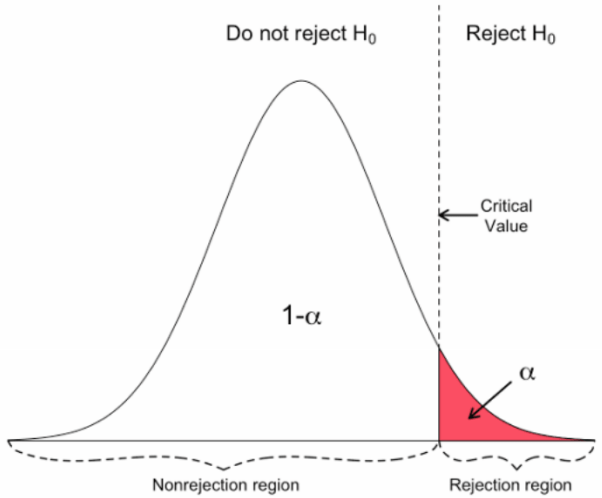
One-sample t-test: location test whether the mean of a population has a value specified in a null hypothesis

```
t_value, p_value = stats.ttest_1samp(samples, mean)
```

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

t-values are correlated to the p-values via the Student's t-distribution, normally calculated simultaneously.

Usual critical value: 0.05 (lower = alternate hypothesis accepted)



Student, "The Probable Error of a Mean", *Biometrika*, **1908**, *6*, 1-25, https://doi.org/10.2307/2331554

Hypothesis testing – t-test (using scipy)

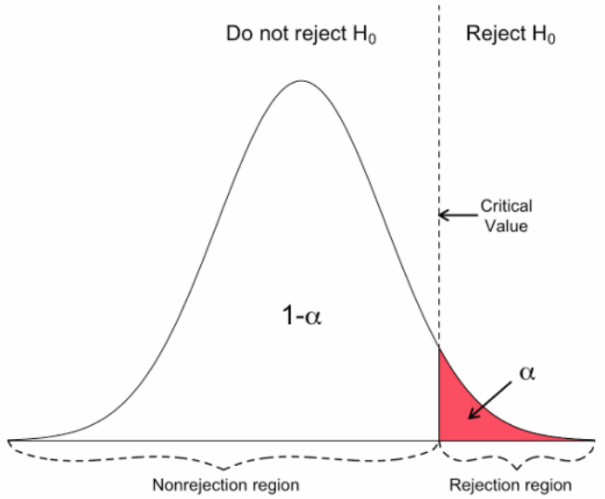
Classic t-test:

Two-sample t-test: A two-sample location test of the null hypothesis which states that the means of two populations are equal.

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

- Assumptions: a) two populations have ca. equal size, b) the variances for the two populations are ca. equal (else: Welsh test)
- Classically: **Unpaired** samples (independent), also variants for **paired** samples (dependent, e.g. repeated measurements)

```
t_value, p_value = stats.ttest_ind(group1, group2, equal_var=True)
```



Usual critical value: 0.05 (lower = alternate hypothesis accepted)

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Acitivity: Hypothesis generation and testing on our data set

Use the curated data set and generate some hypotheses from it.

Test the hypothesis “high lipophilicity means high activity” by applying a t-test.

As a threshold for distinguishing between high and low lipophilicity, try XLogP > 3 as a start (you can change that later and see if it makes a difference)

What is the outcome?